

Learning PGCs over Categorical Variables

1 Representation of the Categorical Mixture Model

We consider a Probabilistic Generating Circuit (PGC) modeling a mixture of r product distributions over m categorical variables. Let each variable have K distinct states. Due to the affine normalization constraint (probabilities summing to 1), each local marginal distribution possesses exactly $D = K - 1$ degrees of freedom.

We designate one state as the baseline (state 0) and represent the remaining D states using indicator variables $\mathbf{x}_j = (x_{j,1}, \dots, x_{j,D}) \in \{0, 1\}^D$, such that $\sum_{s=1}^D x_{j,s} \leq 1$. The normalized local marginal for variable j in component i can be parameterized as an affine function:

$$p(x_j | i) \propto 1 + \sum_{s=1}^D a_{i,j}^{(s)} x_{j,s} \quad (1)$$

where $a_{i,j}^{(s)}$ are the normalized parameters for state s . The generating polynomial for the entire mixture model is:

$$f(\mathbf{x}_1, \dots, \mathbf{x}_m) = \sum_{i=1}^r \mu_i \prod_{j=1}^m \left(1 + \sum_{s=1}^D a_{i,j}^{(s)} x_{j,s} \right) \quad (2)$$

By concatenating these parameters across all m variables and D states, we define the global parameter vector $\alpha_i \in \mathbb{R}^{Dm}$ for the i -th mixture component:

$$\alpha_i = \left[a_{i,1}^{(1)}, \dots, a_{i,1}^{(D)}, a_{i,2}^{(1)}, \dots, a_{i,2}^{(D)}, \dots, a_{i,m}^{(1)}, \dots, a_{i,m}^{(D)} \right]^T \quad (3)$$

2 Generalized Queries for K -ary Models

To perform tensor decomposition, we must populate the second-order ($M_2 \in \mathbb{R}^{Dm \times Dm}$) and third-order ($M_3 \in \mathbb{R}^{Dm \times Dm \times Dm}$) observable moment matrices. The entries of these matrices correspond to the multilinear cross-terms of our hidden parameters across distinct variables.

For a Categorical mixture parameterized by indicators $x_{j,s} \in \{0, 1\}$, the marginalized generating polynomial over any subset of variables expands into a sum of multilinear monomials. Because $x_{j,s} x_{j,t} = 0$ for mutually exclusive states $s \neq t$, no variable can contribute more than degree 1 to any valid monomial.

Let $\mathbf{e}_s \in \{0, 1\}^D$ denote the vector where the s -th component is 1 and all others are 0. Let $\mathbf{0} \in \{0, 1\}^D$ denote the baseline state vector.

2.1 Second-Order Queries (M_2)

To extract the interaction between state s of variable J and state t of variable K ($J \neq K$), we marginalize all other variables by setting them to $\mathbf{0}$, yielding $f_{J,K}(\mathbf{x}_J, \mathbf{x}_K)$.

The expansion of this bivariate polynomial is:

$$f_{J,K}(\mathbf{x}_J, \mathbf{x}_K) = 1 + \sum_{u=1}^D C_J^{(u)} x_{J,u} + \sum_{v=1}^D C_K^{(v)} x_{K,v} + \sum_{u=1}^D \sum_{v=1}^D C_{J,K}^{(u,v)} x_{J,u} x_{K,v} \quad (4)$$

where the bilinear coefficient $C_{J,K}^{(s,t)} = \sum_{i=1}^r \mu_i a_{i,J}^{(s)} a_{i,K}^{(t)}$ is the exact entry we need for M_2 .

To isolate $C_{J,K}^{(s,t)}$, we apply the inclusion-exclusion principle over the specific active states s and t . We define the query as:

$$f_{J,K}(\mathbf{e}_s, \mathbf{e}_t) - f_{J,K}(\mathbf{e}_s, \mathbf{0}) - f_{J,K}(\mathbf{0}, \mathbf{e}_t) + f_{J,K}(\mathbf{0}, \mathbf{0}) \quad (5)$$

Proof of Isolation: Evaluating the four query points:

$$f_{J,K}(\mathbf{e}_s, \mathbf{e}_t) = 1 + C_J^{(s)} + C_K^{(t)} + C_{J,K}^{(s,t)} \quad (6)$$

$$f_{J,K}(\mathbf{e}_s, \mathbf{0}) = 1 + C_J^{(s)} \quad (7)$$

$$f_{J,K}(\mathbf{0}, \mathbf{e}_t) = 1 + C_K^{(t)} \quad (8)$$

$$f_{J,K}(\mathbf{0}, \mathbf{0}) = 1 \quad (9)$$

Substituting these leaves exactly the required bilinear coefficient:

$$\sum_{i=1}^r \mu_i a_{i,J}^{(s)} a_{i,K}^{(t)} \quad (10)$$

Iterating this 4-point query over all $\binom{m}{2}$ variable pairs and all $D \times D$ state combinations fully populates the off-diagonal blocks of M_2 .

2.2 Third-Order Queries (M_3)

To extract the third-order interaction across variables J, K , and L for states s, t , and u respectively, we marginalize to $f_{J,K,L}(\mathbf{x}_J, \mathbf{x}_K, \mathbf{x}_L)$ and the query follows similarly :-

$$\begin{aligned} & f_{J,K,L}(\mathbf{e}_s, \mathbf{e}_t, \mathbf{e}_u) \\ & - f_{J,K,L}(\mathbf{e}_s, \mathbf{e}_t, \mathbf{0}) - f_{J,K,L}(\mathbf{e}_s, \mathbf{0}, \mathbf{e}_u) - f_{J,K,L}(\mathbf{0}, \mathbf{e}_t, \mathbf{e}_u) \\ & + f_{J,K,L}(\mathbf{e}_s, \mathbf{0}, \mathbf{0}) + f_{J,K,L}(\mathbf{0}, \mathbf{e}_t, \mathbf{0}) + f_{J,K,L}(\mathbf{0}, \mathbf{0}, \mathbf{e}_u) \\ & - f_{J,K,L}(\mathbf{0}, \mathbf{0}, \mathbf{0}) \end{aligned} \quad (11)$$

By identical inclusion-exclusion logic, this 8-point query perfectly annihilates all baseline, linear, and bilinear terms, leaving strictly the trilinear cross-term:

$$\sum_{i=1}^r \mu_i a_{i,J}^{(s)} a_{i,K}^{(t)} a_{i,L}^{(u)} \quad (12)$$

By iterating this query over all variable triplets $J < K < L$ and all $D \times D \times D$ state combinations, we fully populate the off-diagonal components of the third-order tensor M_3 .

3 Identifiability and Recovery of Diagonal Moment Entries

We achieve this by exploiting the low-rank structure of the global moment matrix. Recall that the complete second-order moment matrix can be expressed as the outer product $M_2 = A \text{Diag}(\mu) A^T$. Because $\text{Diag}(\mu)$ is an $r \times r$ matrix, it immediately follows that:

$$\text{rank}(M_2) \leq r \quad (13)$$

provided $m \geq 2r + 1$. We can leverage this rank boundary to isolate the unknown diagonal entries using strictly observable off-diagonal blocks.

4 Small Restriction and Jacobian Rank

Choose an *alive set* of size m . For any degree tuple $E \in \{0, 1, \dots, D\}^m$, the coefficient of the corresponding monomial is given by:

$$c_E = \sum_{i=1}^r \mu_i \prod_{s=1}^D \prod_{j: E_j=s} a_{i,j}^{(s)} \quad (14)$$

We form the Jacobian matrix J_r with rows indexed by E and columns indexed by the parameters $\{\mu_i\} \cup \{a_{i,j}^{(s)}\}$. Let M_r denote the submatrix restricted to the parameter columns and rows $E \neq \mathbf{0}$. M_r has exactly Dmr columns. We will show $\text{rank}(M_r) = Dmr$.

We specialize to the witness assignment $a_{i,j}^{(s)} = \theta_i^{(s)} \cdot u_j^{(s)}$ for all $i \in [r], j \in [m], s \in [D]$. We systematically isolate the parameters by restricting our attention to pure degree tuples.

For each state $s \in \{1, \dots, D\}$, let $\mathcal{E}_k^{(s)} = \{E \in \{0, \dots, D\}^m : |\{j : E_j = s\}| = k, \text{ and } E_j = 0 \text{ otherwise}\}$ be the set of tuples containing exactly k instances of state s and $m - k$ instances of the baseline state.

For any $E \in \mathcal{E}_k^{(s)}$, the indicator for any other state $t \neq s$ is identically zero. Therefore, all partial derivatives with respect to $a_{i,\ell}^{(t)}$ vanish entirely. The remaining non-zero derivatives with respect to state s simplify to:

$$\frac{\partial c_E}{\partial a_{i,\ell}^{(s)}} = \mu_i (\theta_i^{(s)})^{k-1} \cdot \mathbf{1}_{\{E_\ell=s\}} \prod_{\substack{j \neq \ell \\ E_j=s}} u_j^{(s)} \quad (15)$$

Notice this structure is perfectly identical to the binary setting. Thus, grouping the rows of M_r indexed by $\mathcal{E}_k^{(s)}$ yields a block-row $M_{\mathcal{E}_k^{(s)}}$ where only the columns corresponding to state s are non-zero:

$$M_{\mathcal{E}_k^{(s)}} = \left(\mathbf{0} \quad \dots \quad \left| \quad \mu_1 (\theta_1^{(s)})^{k-1} G_k^{(s)} \quad \dots \quad \mu_r (\theta_r^{(s)})^{k-1} G_k^{(s)} \quad \right| \quad \dots \quad \mathbf{0} \right) \quad (16)$$

where $G_k^{(s)}$ is the standard $\binom{m}{k} \times m$ binary incidence matrix.

Reducing to a decoupled Vandermonde structure. By Lemma C.2 of the binary setting, for each $k \in \{1, \dots, m-1\}$, the incidence matrix $G_k^{(s)}$ has full column rank m , allowing row operations R_k to extract the identity matrix I_m .

Applying these operations independently to the pure blocks for all states $s \in [D]$, the submatrix M' of M_r completely decouples into a block-diagonal matrix of D independent parameter spaces:

$$V = \begin{pmatrix} \left(A_{r \times r}^{(1)} \otimes I_m \right) D_\mu^{(1)} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \left(A_{r \times r}^{(2)} \otimes I_m \right) D_\mu^{(2)} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \left(A_{r \times r}^{(D)} \otimes I_m \right) D_\mu^{(D)} \end{pmatrix} \quad (17)$$

where $A_{r \times r}^{(s)}$ are standard Vandermonde matrices generated by the witness parameters $\theta_i^{(s)}$.

Because the determinant of a block-diagonal matrix is the product of its diagonal blocks, and each Vandermonde block $A^{(s)}$ is non-singular for generically chosen distinct $\theta_i^{(s)}$, every block has full rank mr . Summing across the D blocks, $\text{rank}(V) = \sum_{s=1}^D mr = Dmr$.

Because V is a submatrix of M_r , we have $\text{rank}(M_r) \geq Dmr$. Since M_r has exactly Dmr columns, its rank is strictly Dmr .

Separating r from $r - 1$. We require the minimum rank of the r -mixture to strictly exceed the maximum rank of the $(r - 1)$ -mixture:

$$Dmr + 1 > (r - 1) + Dm(r - 1) \implies Dm > r - 2 \quad (18)$$

Since we extracted r blocks for each state ($k = 1, \dots, r$), we require $m - 1 \geq r \implies 2m \geq 2r + 2$. Since $D \geq 1$ for any non-trivial categorical variable, $Dm \geq m > r - 2$ is always satisfied.

Proposition 3.1. *For $m \geq r + 1$, in the generic setting, restricting to any alive set of size m preserves the mixture size r for a K -ary categorical Probabilistic Generating Circuit.*